



Self-Reflection: Work with datasets

Activity overview

Recently, you have explored formulas and functions in spreadsheets, created custom data tables, and explored how to save time with structured thinking. Now, you'll work with and manipulate datasets, then respond to brief questions.

These things will help you to develop insights into your own learning journey and prepare you to apply your knowledge of analytics, crafting SMART questions, and developing a scope of work (SOW). As you answer questions—and come up with questions of your own—consider the concepts, practices, and principles that have helped refine your understanding and reinforce your learning. You've done the hard work, so make sure to get the most out of it by practicing what you've learned!

Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the question at the end of the activity before going to the next course item.

Step 1: Access the template

To use the template for this course item, click the link below and select “Use Template.”

Link to template: [Data Analysis Project Scope-of-Work \(SOW\) Template](#)

OR

If you don't have a Google account, download the template directly from the attachment below.

[Scope-Of-Work Template](#)

[DOCX File](#)

Step 2: Examine a dataset on Kaggle

Kaggle is a data science platform that houses a variety of datasets. You will need to create a free account to begin exploring and analyzing different datasets. Then, this activity will provide an example of how to use a dataset to formulate SMART questions, create an SOW, and perform analytics in a spreadsheet. After that, you'll have an opportunity to explore a dataset of your choice to formulate your own SMART questions, create your own SOW, and perform analytics in a spreadsheet that you can add to your professional portfolio.

1. Download [Lettuce Growth Days](#) from the Kaggle website. After you extract the files, open the lettuce_dataset CSV file in Google Sheets or Excel.
2. Analyze the dataset's column names, values, and summary statistics (mean, median, quartiles, and min/max values), as well as contemplate goals to formulate SMART questions. SMART questions for the Lettuce Growth Days data could include:
 - a. **Specific:** Is there a connection between humidity and temperature?
 - b. **Measurable:** How much growth is normal or expected over a one-week period?
 - c. **Action-oriented:** If total dissolved solids are increased when growing lettuce, will it shorten the time needed for the plants to reach maturity?
 - d. **Relevant:** Based on the dataset, how can predictions about growth be made given the conditions?
 - e. **Time-bound:** What is the date range for this dataset? Do we need an understanding of this data for next season?
3. Based on the SMART questions, create an SOW that outlines the analytics tasks. An example of an SOW for Lettuce Growth Days can be found in the [Lettuce dataframe: Example SOW](#).
4. Perform basic analytics on the dataset. For the Lettuce Growth Days dataset, analytics could include:
 - a. Converting Celsius to Fahrenheit

- b. Converting days to weeks
- c. Modifying the humidity column
- d. Creating a new spreadsheet to perform exploratory data analysis

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New Notebook

 Download (28 kB)



Lettuce Growth Days

Prediction of Growth Days of Lettuce

Data Card

Code (3)

Discussion (0)

About Dataset

Usability ⓘ
10.00

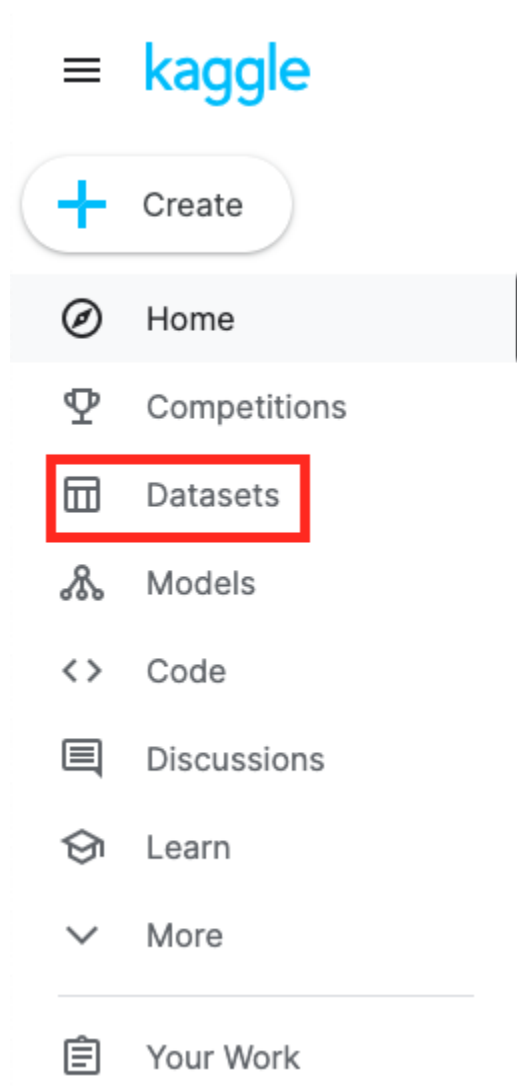


Step 3: Explore more datasets

1. From the [Kaggle](#) homepage, select **Datasets** to view the available datasets.
2. Select a category that interests you or that aligns with your career goals.
Select **Download** to download the data and begin analyzing it. Based on the dataset you chose, create a list of 3-5 SMART questions, develop a completed SOW, and create a new Google Sheet with performed analytics that best align to your dataset.

Note: It's important to check the Kaggle Usability Score when choosing a dataset on Kaggle; the higher the score, the easier the data will be to work with. Find the Usability Score in the dataset information. It ranks the dataset's completeness, credibility, and compatibility.

3. Share your findings, SOW, and analytics in the [Google Data Analytics Community](#).



Usability
10.00

License
Other (specific)

Expected updates
Weekly

Tags
Food
Asia

This score is calculated by Kaggle.

Completeness · 100%

- ✓ Subtitle
- ✓ Tag
- ✓ Description
- ✓ Cover Image

Credibility · 100%

- ✓ Source/Provenance
- ✓ Public Notebook
- ✓ Update Frequency

Compatibility · 100%

- ✓ License
- ✓ File Format
- ✓ File Description
- ✓ Column Description

Pro Tip: Save the blank activity template

Be sure to save a blank copy of the template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

Consider what you've learned about working with datasets to formulate SMART questions, crafting an SOW, and performing data analytics in this reflection:

- Why is it important to create SMART questions about your datasets? How do these questions benefit your work as a data professional?
- Why is it important to determine your SMART questions and answers before crafting an SOW?
- Why is it important to perform data analysis on datasets?

Now, write 2-3 sentences (40-60 words) in response to each of these questions. Enter your response in the text box below.

Answer:

Creating SMART (Specific, Measurable, Achievable, Relevant, Time-bound) questions brings clarity and focus to data analysis. They ensure that your investigation has a defined purpose and boundaries, preventing aimless exploration and maximizing the efficiency of your work.

SMART questions guide the selection of appropriate data, analytical techniques, and visualizations. This leads to more targeted and insightful findings, enabling you to provide actionable recommendations to stakeholders and demonstrate the value of data-driven decision-making.

Defining SMART questions and anticipated answers upfront provides a clear understanding of the project's objectives and expected outcomes. This clarity ensures the scope of work accurately reflects the necessary tasks, resources, and timelines required to achieve those specific, measurable goals.

Data analysis is crucial for extracting meaningful insights and patterns hidden within raw data. It transforms complex information into understandable knowledge, enabling informed decision-making, problem-solving, and the identification of new opportunities across various domains.

Correct

Great work reinforcing your learning! A thoughtful reflection on this topic would highlight the importance of using SMART questions to help craft an SOW. In addition, your response should include why an SOW is important to develop for a project and how your data analysis drove you to your completed SOW.

The SOW is mostly an organizational tool to help analysts make effective and impactful decisions. It's about being technical by taking the same steps each time to make sure the analyst acts with intention. Being able to describe your process and the methods highlights your ability to be thorough, follow through on projects, and apply effective frameworks to your work.